

OPENCV

From Webcam to Computer Vision Projects

"Your webcam can become more than just a camera."

DEFINITION

OpenCV = Open Source Computer Vision Library



Works with Images

Process, transform, and analyze static visual files.



Works with Videos

Frame-by-frame decoding and automated analysis.



Webcam Stream Input

Real-time feed capture directly from physical sensors.



Visual Understanding

Helps computers interpret matrix pixel data meaningfully.

SYSTEM ARCHITECTURE FLOW



CODETOCRACK NOTE

OpenCV is one of the best starting points if you want to move from standard Python coding into Computer Vision.

WHAT YOU NEED

Python

NumPy

OpenCV



Image Processing

Resize • Crop • Rotate • Flip • Filters



Webcam Projects

Real-time camera processing feed



Color Detection

Detect specific colors from a camera



Hand Tracking

Track hand and finger movements



Air Drawing

Draw on screen using your finger



Face Detection

Detect human faces in real time



Eye Detection

Detect & track gaze direction



OCR Extraction

Convert image text into string data



Document Scanner

Scan & perspective-transform docs



Object Detection

Detect items inside images/videos



Object Tracking

Track trajectory of moving targets



People Counting

Count people entering/exiting zones



Vehicle Detection

Detect & count road vehicles



Motion Detection

Detect movement changes via camera



Smart CCTV

Build intelligent security systems



Lane Detection

Detect road lanes from vehicle video

Endless Real-World Applications

Combine basic image filters to create autonomous AI systems.

[LEARN BY BUILDING](#)

LEVEL 01

IMAGE BASICS

- ✓ Read, Display & Save Image
- ✓ Image Dimensions & Pixels
- ✓ Resize & Crop Operations
- ✓ Rotate & Flip Manipulations

IMAGE → PROCESS → OUTPUT

LEVEL 02

IMAGE PROCESSING

- ✓ Grayscale, BGR / RGB & HSV
- ✓ GaussianBlur & Thresholding
- ✓ Canny Edge & Contour Detection
- ✓ Shape Detection & Annotations

RAW IMAGE → PROCESSING → DETECTED RESULT

LEVEL 03

REAL-TIME COMPUTER VISION

- ✓ Webcam Video Capture Loop
- ✓ Frame-by-Frame Processing
- ✓ Real-Time Color & Motion Detection
- ✓ Face & Object Tracking Pipelines

WEBCAM → FRAME → OPENCV → RESULT

LEVEL 04

AI + COMPUTER VISION INTEGRATION

OpenCV works seamlessly alongside modern neural network stacks to power state-of-the-art vision models:

OpenCV + MediaPipe

OpenCV + YOLO

OpenCV + OCR

OpenCV + Custom AI

BEGINNER

- ✓ Image Editor
- ✓ Grayscale Converter
- ✓ Image Resizer
- ✓ Color Detector
- ✓ Edge Detector
- ✓ Shape Detector
- ✓ Webcam Filter
- ✓ Motion Detector

INTERMEDIATE

- ✓ Face Detection
- ✓ Eye Detection
- ✓ Document Scanner
- ✓ OCR Scanner
- ✓ Object Tracking
- ✓ People Counter
- ✓ Vehicle Counter
- ✓ Virtual Drawing Board

ADVANCED

- ✓ Air Mouse
- ✓ Gesture Calculator
- ✓ Gesture Volume Control
- ✓ Smart CCTV
- ✓ Number Plate Detection
- ✓ Lane Detection
- ✓ Real-Time Object Detect
- ✓ AI Vision Applications

CODETOCRACK PROJECT IDEA

FLAGSHIP BUILD

“Control your computer without touching the mouse.”

HAND



CAMERA



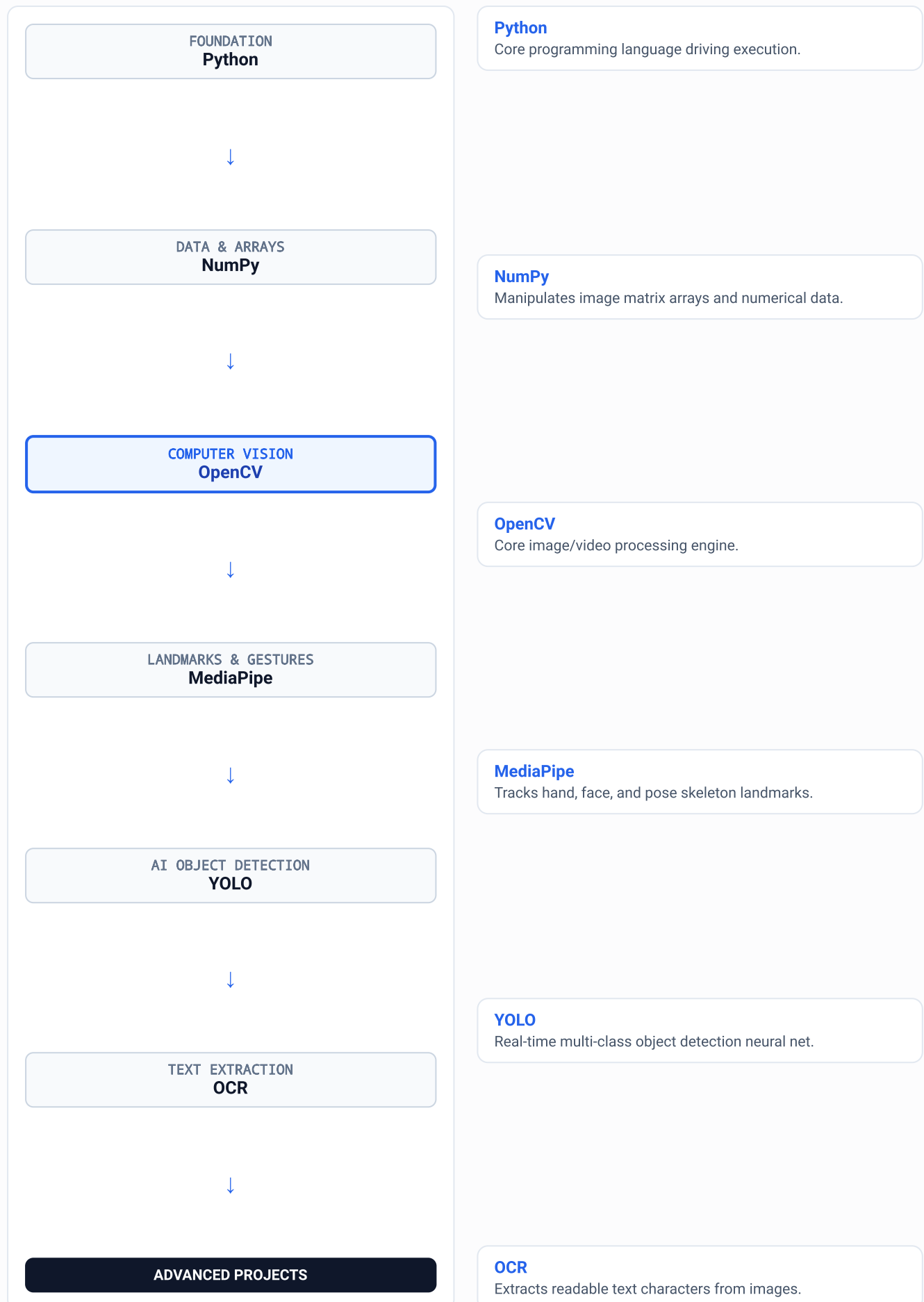
OPENCV



GESTURE



ACTION



01 Python Basics	02 NumPy Basics	03 OpenCV Intro
04 Webcam Stream	05 MediaPipe/YOLO	06 Build Projects

IMPORTANT FUNCTIONS CHEAT SHEET

cv2.imread()	cv2.imshow()
cv2.imwrite()	cv2.resize()
cv2.cvtColor()	cv2.GaussianBlur()
cv2.threshold()	cv2.Canny()
cv2.rectangle()	cv2.circle()
cv2.putText()	cv2.VideoCapture()
cv2.findContours()	cv2.inRange()

DON'T JUST LEARN OPENCV. BUILD WITH IT.

Start with images • Move to webcam • Learn detection • Add AI